

# Household Food Security and Self Rated General Health Among United States Adults

A Survey Weighted Analysis of the 2024 National Health Interview Survey

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|                                       |                                                        |                                             |
|---------------------------------------|--------------------------------------------------------|---------------------------------------------|
| <b>30,230</b><br>complete-case adults | <b>2.60</b><br>adjusted POR for very low food security | <b>2.22–3.04</b><br>95% confidence interval |
|---------------------------------------|--------------------------------------------------------|---------------------------------------------|

Public-use secondary analysis • Reproducible R workflow • Complex survey design accounted for

# Abstract

**Background.** Food insecurity is associated with adverse health outcomes, but recent national evidence that preserves the full ordering of both food security and self-rated health remains useful for surveillance and policy research.

**Methods.** This cross-sectional analysis used the 2024 National Health Interview Survey Sample Adult public-use file. The analytic cohort included 30,230 adults with complete data on self-rated health, four-category household food security, and prespecified covariates. Estimates incorporated the final annual sample weight, strata, and primary sampling units. Survey-weighted cumulative-logit models produced proportional odds ratios (PORs) for worse self-rated health.

**Results.** The survey-weighted prevalence of fair or poor health was 12.1% among adults with high food security and 35.3% among adults with very low food security. Relative to high food security, adjusted odds of worse health were higher for marginal food security (POR 1.89, 95% CI 1.70–2.11), low food security (POR 1.98, 95% CI 1.73–2.26), and very low food security (POR 2.60, 95% CI 2.22–3.04; overall design-based  $p < 0.001$ ). Threshold-specific models remained directionally consistent, although effect magnitudes varied across health cut points.

**Conclusions.** Lower household food security showed a strong graded association with worse self-rated health after adjustment for demographic, socioeconomic, behavioral, and health-related characteristics. Because the data are cross-sectional and self-reported, the results describe association rather than causation.

**Keywords:** *food security; self-rated health; NHIS; ordinal logistic regression; complex survey analysis*

## Project at a glance

- Population: 30,230 adults representing approximately 240 million U.S. adults after survey weighting.
- Exposure: high, marginal, low, and very low household food security during the past 30 days.
- Outcome: self-rated general health ordered from excellent to poor.
- Method: survey-weighted proportional-odds regression with design-based standard errors and prespecified sensitivity checks.
- Main finding: very low food security was associated with 2.60 times the adjusted odds of reporting worse health compared with high food security.

### Interpretation guardrail

A proportional odds ratio is a cumulative odds measure, not a risk ratio. The cross-sectional design does not establish that food insecurity caused poorer health.

# Introduction

Food security exists when households have dependable access to enough food for an active, healthy life. In 2024, 13.7% of U.S. households were food insecure at some point during the year, including 5.4% that experienced very low food security [1]. Food insecurity is therefore both a material hardship and a population-health concern. It can constrain diet quality, increase stress, and force trade-offs among food, housing, medication, and other essential expenses.

Self-rated general health is a concise measure that integrates physical, mental, and functional dimensions of health. Although subjective, it is consistently associated with later morbidity and mortality [3]. Previous studies have linked food insecurity with poorer self-reported health in U.S. adults and in populations facing disability or other forms of social disadvantage [4–7]. Much of this literature, however, dichotomizes health or food security, uses regional samples, or does not fully account for a survey's complex design.

This analysis examined whether household food security status was associated with five-level self-rated general health among U.S. adults in the 2024 National Health Interview Survey (NHIS). The primary hypothesis was that progressively lower food security would be associated with progressively higher odds of worse self-rated health, even after adjustment for demographic, socioeconomic, behavioral, and health-related characteristics. A secondary objective was to evaluate whether the direction and magnitude of the association were reasonably consistent across cumulative health thresholds.

## Methods

### Study design and data source

The analysis used the 2024 NHIS Sample Adult public-use file released by the National Center for Health Statistics (NCHS). NHIS is a cross-sectional household interview survey designed to represent the civilian, noninstitutionalized U.S. population [2]. The 2024 Sample Adult file contains 32,629 completed adult interviews; the reported Sample Adult response rate was 47.9% [2]. Survey documentation, questionnaires, codebooks, and the public-use data are available from NCHS.

### Study population

Adults aged 18 years or older were eligible. Values coded as refused, not ascertained, do not know, or unknown were treated as missing. A single complete-case cohort was used for descriptive and regression analyses so that all reported tables refer to the same population. Of 32,629 sample adults, 2,399 were excluded because at least one model variable was missing, leaving 30,230 adults (92.6% of the public-use sample). Variable-specific missingness is reported in Supplementary Table S4.

### Outcome

Self-rated general health was measured with PHSTAT\_A. Responses were ordered as excellent, very good, good, fair, and poor. The order was coded from better to worse health, so a POR greater than 1 indicates higher cumulative odds of being in a worse health category.

### Primary exposure

Household food security was measured with FDSCAT4\_A, a four-category recode based on household food-security questions referring to the past 30 days. Categories were high, marginal, low, and very low food security. High food security was the reference category.

### Covariates

Covariates were selected before model fitting based on the project's conceptual framework and the original course analysis. They included age group, sex, race and ethnicity, educational attainment, marital status, U.S. Census region, cigarette-smoking status, alcohol-drinking status, leisure-time physical activity, body mass index

category, sleep duration, and health-insurance coverage. Detailed source variables, category definitions, and reference groups appear in Supplementary Table S1.

The combined race and Hispanic-origin variable HISPALLP\_A was used directly; it was not cross-classified with a separate race variable. Educational attainment was grouped consistently across all analyses. Alcohol codes 2–4 were classified as former drinking and codes 5–9 as current drinking. Extreme but valid activity values were retained as evidence of any activity, values indicating inability were treated as no activity for that component, and refused or unknown activity responses were treated as missing.

## **Survey design and statistical analysis**

All population estimates and regression models incorporated the final annual sample-adult weight (WTFA\_A), public-use pseudo-strata (PSTRAT), and public-use primary sampling units (PPSU), with strata nested as recommended by NCHS [2]. The survey design was defined on the full Sample Adult file and then subset to complete cases, preserving the design information needed for valid variance estimation. Descriptive results are presented as unweighted counts with survey-weighted percentages; age is shown as a survey-weighted mean with standard error (SE).

Cumulative-logit proportional-odds models were fit using survey-weighted ordinal logistic regression [8,10]. The unadjusted model included food security only. The adjusted model included food security and all prespecified covariates. Estimates are reported as PORs with 95% confidence intervals based on the model's design degrees of freedom. The overall contribution of food security was assessed with a design-based Wald F test. Two-sided p values below 0.05 were considered statistically significant; emphasis was placed on effect size and precision rather than significance alone.

## **Diagnostics and sensitivity analyses**

Because a single POR assumes that each predictor has the same cumulative effect across outcome cut points, four survey-weighted binary logistic models were fit at successive health thresholds. These models do not replace a formal design-based global proportional-odds test, but they show whether the primary exposure retains the same direction and whether its magnitude is stable across thresholds. Coefficient-level weighted variance inflation factors (VIFs) were calculated from the adjusted design matrix to assess collinearity. Complete-case retention and variable-specific missingness were documented.

## **Software and reproducibility**

The analysis was performed in R 4.3.3 using survey 4.2-1 and MASS 7.3-60. The workflow reads the official NCHS CSV file, applies explicit recodes, constructs the survey design, fits the models, and writes machine-readable tables and figures. The source dataset is not redistributed; the accompanying repository should direct users to the official NCHS download page and provide the expected local file path.

# Results

## Analytic sample

The complete-case cohort included 30,230 adults and represented approximately 240 million U.S. adults. Unweighted counts were 25,715 with high food security, 1,883 with marginal food security, 1,495 with low food security, and 1,137 with very low food security. Survey-weighted shares were 83.9%, 6.8%, 5.5%, and 3.7%, respectively.

## Weighted characteristics

The weighted mean age was 48.3 years (SE 0.17). Lower-food-security groups were younger on average and included larger weighted proportions of women, non-Hispanic Black adults, Hispanic adults, adults with less education, current smokers, adults with obesity, adults sleeping fewer than seven hours, and adults without health coverage. For example, the weighted prevalence of current smoking was 8.1% in the high-food-security group and 24.4% in the very-low-food-security group. The corresponding weighted prevalence of obesity was 31.9% and 45.8% (Table 1).

**Table 1.** *Characteristics of U.S. adults overall and by household food-security status*

| Characteristic                             | Overall       | High          | Marginal     | Low         | Very low    |
|--------------------------------------------|---------------|---------------|--------------|-------------|-------------|
| <b>Age</b>                                 |               |               |              |             |             |
| Weighted mean (SE), years                  | 48.3 (0.17)   | 49.2 (0.18)   | 43.7 (0.46)  | 44.6 (0.61) | 41.7 (0.56) |
| 18-34                                      | 6,241 (28.9)  | 4,993 (27.6)  | 544 (36.3)   | 377 (34.0)  | 327 (38.0)  |
| 35-49                                      | 6,685 (24.6)  | 5,482 (23.9)  | 481 (27.7)   | 389 (27.5)  | 333 (30.4)  |
| 50-64                                      | 7,169 (23.6)  | 6,027 (23.8)  | 449 (22.1)   | 398 (23.6)  | 295 (21.8)  |
| 65 or older                                | 10,135 (22.9) | 9,213 (24.8)  | 409 (13.9)   | 331 (14.9)  | 182 (9.8)   |
| <b>Sex</b>                                 |               |               |              |             |             |
| Female                                     | 16,236 (51.0) | 13,528 (49.8) | 1,098 (54.6) | 933 (60.2)  | 677 (58.1)  |
| <b>Race and ethnicity</b>                  |               |               |              |             |             |
| Non-Hispanic White                         | 20,382 (61.9) | 18,060 (64.9) | 990 (47.4)   | 705 (41.5)  | 627 (51.1)  |
| Non-Hispanic Black                         | 2,933 (11.4)  | 2,151 (9.9)   | 286 (16.7)   | 302 (22.4)  | 194 (20.2)  |
| Hispanic                                   | 4,327 (17.7)  | 3,280 (15.9)  | 458 (28.5)   | 369 (28.9)  | 220 (20.7)  |
| Non-Hispanic Asian                         | 1,721 (6.2)   | 1,592 (6.8)   | 67 (4.1)     | 39 (2.4)    | 23 (2.1)    |
| Non-Hispanic AIAN, other, or multiple race | 867 (2.8)     | 632 (2.5)     | 82 (3.3)     | 80 (4.9)    | 73 (5.9)    |
| <b>Educational attainment</b>              |               |               |              |             |             |
| Bachelor's degree or higher                | 11,661 (34.1) | 10,907 (38.0) | 369 (16.7)   | 219 (11.9)  | 166 (12.1)  |
| Some college or associate degree           | 8,456 (29.1)  | 7,027 (28.7)  | 604 (32.9)   | 479 (29.5)  | 346 (30.9)  |
| High school graduate or GED                | 7,708 (26.7)  | 6,093 (25.0)  | 651 (33.7)   | 519 (35.4)  | 445 (39.9)  |
| Less than high school                      | 2,405 (10.0)  | 1,688 (8.3)   | 259 (16.7)   | 278 (23.1)  | 180 (17.1)  |
| <b>Marital status</b>                      |               |               |              |             |             |
| Married                                    | 13,972 (52.6) | 12,602 (55.5) | 658 (41.4)   | 450 (37.6)  | 262 (29.0)  |
| Living with partner                        | 1,958 (8.6)   | 1,532 (7.9)   | 193 (12.4)   | 134 (11.8)  | 99 (12.8)   |
| Neither married nor living with partner    | 14,300 (38.8) | 11,581 (36.6) | 1,032 (46.2) | 911 (50.6)  | 776 (58.2)  |
| <b>U.S. Census region</b>                  |               |               |              |             |             |

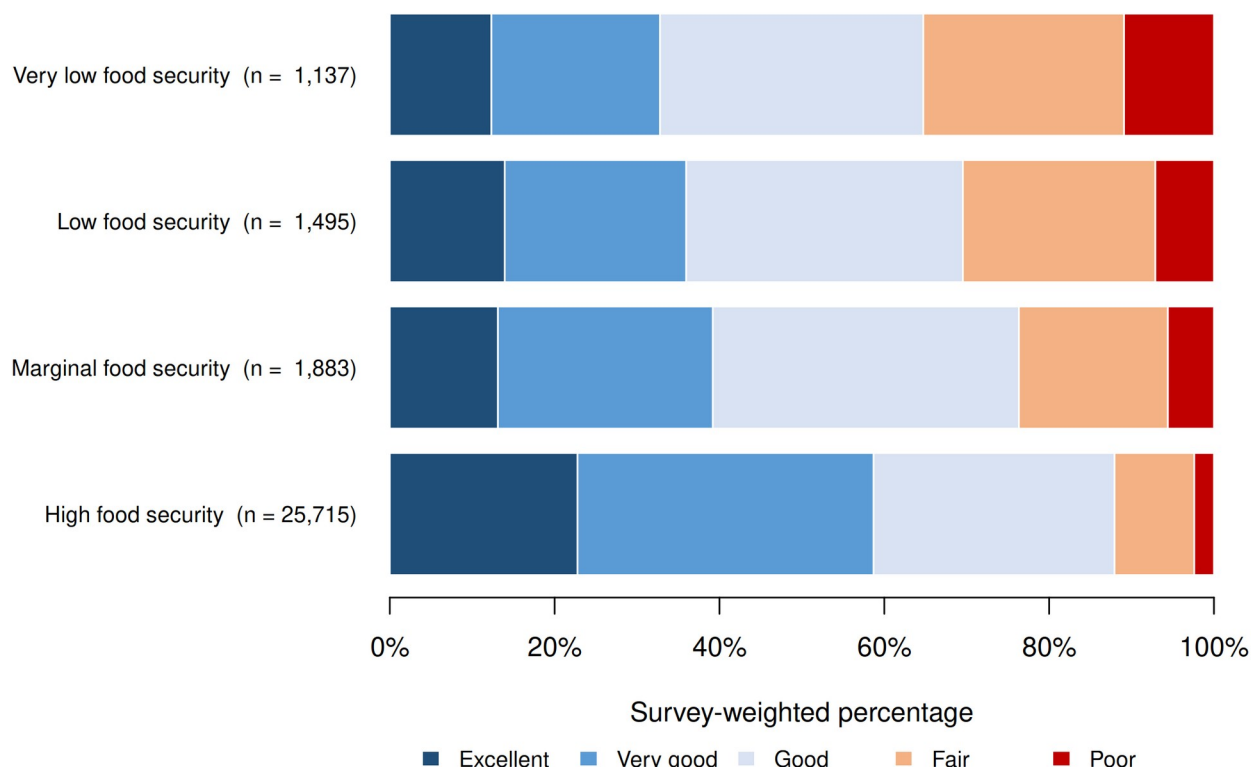
| Characteristic                                | Overall       | High          | Marginal     | Low        | Very low   |
|-----------------------------------------------|---------------|---------------|--------------|------------|------------|
| Northeast                                     | 4,669 (16.9)  | 4,062 (17.3)  | 270 (15.5)   | 190 (14.2) | 147 (15.5) |
| Midwest                                       | 7,019 (20.7)  | 6,069 (21.3)  | 382 (17.8)   | 303 (16.9) | 265 (20.2) |
| South                                         | 10,937 (38.7) | 9,060 (37.5)  | 758 (42.4)   | 662 (49.2) | 457 (43.9) |
| West                                          | 7,605 (23.7)  | 6,524 (24.0)  | 473 (24.3)   | 340 (19.8) | 268 (20.4) |
| <b>Smoking status</b>                         |               |               |              |            |            |
| Never smoker                                  | 19,275 (67.6) | 16,784 (68.9) | 1,114 (64.3) | 831 (60.4) | 546 (55.2) |
| Former smoker                                 | 7,776 (22.5)  | 6,710 (23.0)  | 463 (21.0)   | 326 (19.1) | 277 (20.4) |
| Current smoker                                | 3,179 (9.9)   | 2,221 (8.1)   | 306 (14.7)   | 338 (20.5) | 314 (24.4) |
| <b>Alcohol status</b>                         |               |               |              |            |            |
| Lifetime abstainer                            | 3,402 (13.5)  | 2,787 (12.9)  | 260 (15.9)   | 231 (18.6) | 124 (13.2) |
| Former drinker                                | 6,232 (18.0)  | 5,131 (17.3)  | 436 (20.5)   | 360 (20.5) | 305 (24.0) |
| Current drinker                               | 20,596 (68.6) | 17,797 (69.7) | 1,187 (63.6) | 904 (60.9) | 708 (62.8) |
| <b>Physical activity</b>                      |               |               |              |            |            |
| No moderate or vigorous leisure-time activity | 7,844 (25.5)  | 6,188 (23.5)  | 613 (32.2)   | 604 (40.0) | 439 (37.7) |
| <b>Body mass index</b>                        |               |               |              |            |            |
| Healthy weight                                | 9,395 (31.1)  | 8,286 (32.3)  | 472 (24.8)   | 361 (24.6) | 276 (25.1) |
| Underweight                                   | 518 (1.7)     | 424 (1.6)     | 38 (2.1)     | 27 (1.8)   | 29 (3.4)   |
| Overweight                                    | 10,176 (33.5) | 8,837 (34.2)  | 594 (32.2)   | 448 (30.6) | 297 (25.7) |
| Obesity                                       | 10,141 (33.6) | 8,168 (31.9)  | 779 (40.9)   | 659 (43.0) | 535 (45.8) |
| <b>Sleep duration</b>                         |               |               |              |            |            |
| 7-9 hours                                     | 19,800 (65.8) | 17,436 (68.1) | 1,063 (57.1) | 805 (55.5) | 496 (44.4) |
| Less than 7 hours                             | 9,266 (30.4)  | 7,367 (28.4)  | 720 (37.8)   | 603 (38.4) | 576 (49.5) |
| More than 9 hours                             | 1,164 (3.8)   | 912 (3.4)     | 100 (5.1)    | 87 (6.1)   | 65 (6.1)   |
| <b>Health insurance</b>                       |               |               |              |            |            |
| Not covered                                   | 2,104 (8.9)   | 1,494 (7.4)   | 220 (14.6)   | 210 (18.4) | 180 (19.4) |

**Note.** Values are unweighted n (survey-weighted %) unless otherwise indicated. All columns use the same complete-case cohort. Percentages may not total 100 because of rounding. AIAN = American Indian or Alaska Native; GED = General Educational Development; SE = standard error.

## Self-rated health across food-security groups

Self-rated health shifted toward worse categories as food security decreased. The combined weighted prevalence of fair or poor health was 12.1% among adults with high food security, 23.7% among those with marginal food security, 30.5% among those with low food security, and 35.3% among those with very low food security (Figure 1). Poor health alone was reported by a weighted 2.4% of the high-food-security group and 10.9% of the very-low-food-security group.

**Figure 1.** Survey-weighted distribution of self-rated general health by household food-security status



**Note.** Percentages incorporate WTFA\_A, PSTRA1, and PPSU. Group sample sizes shown in the figure are unweighted.

## Ordinal regression results

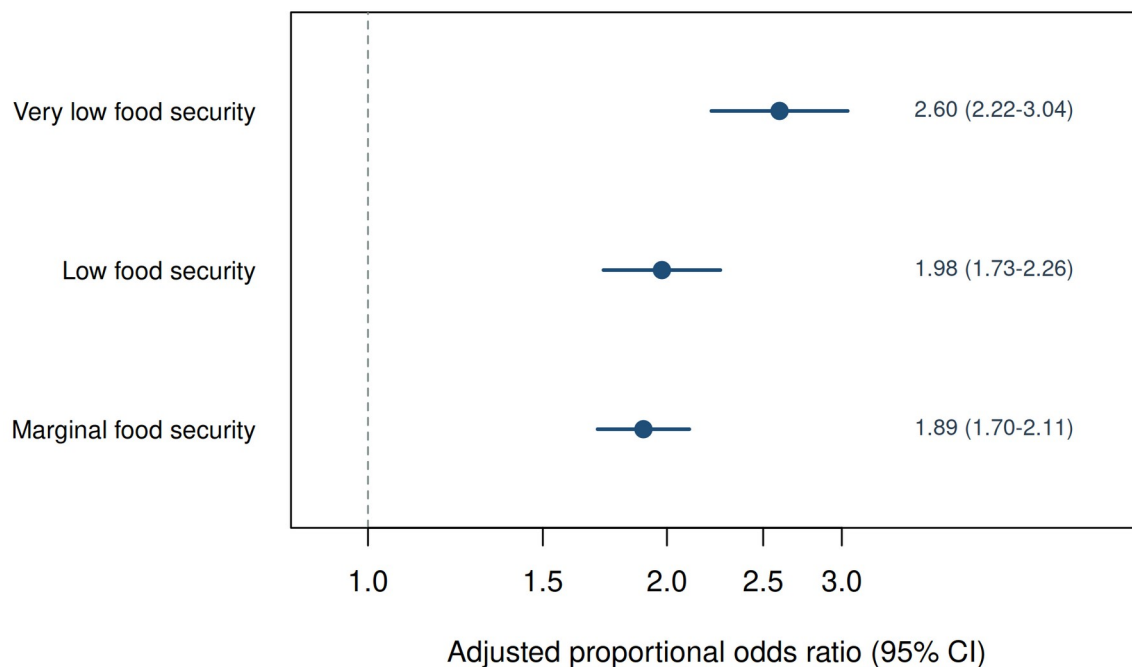
In the unadjusted model, the odds of worse health increased across lower food-security categories. Adjustment attenuated the estimates but did not remove the association. Relative to high food security, the adjusted POR was 1.89 (95% CI 1.70–2.11) for marginal food security, 1.98 (95% CI 1.73–2.26) for low food security, and 2.60 (95% CI 2.22–3.04) for very low food security (Table 2 and Figure 2). The overall association was statistically significant in both models (design-based Wald  $p < 0.001$ ).

**Table 2.** Association between household food security and worse self-rated general health

| Food-security category | Unadjusted POR (95% CI) | p value | Adjusted POR (95% CI) | p value |
|------------------------|-------------------------|---------|-----------------------|---------|
| Marginal food security | 2.16 (1.94–2.40)        | <0.001  | 1.89 (1.70–2.11)      | <0.001  |
| Low food security      | 2.64 (2.30–3.03)        | <0.001  | 1.98 (1.73–2.26)      | <0.001  |
| Very low food security | 3.31 (2.84–3.86)        | <0.001  | 2.60 (2.22–3.04)      | <0.001  |

**Note.** Reference = high food security. POR = proportional odds ratio; CI = confidence interval. POR > 1 indicates higher cumulative odds of worse health. Overall design-based Wald F tests: unadjusted  $F(3, 607) = 178.90$ ,  $p < 0.001$ ; adjusted  $F(3, 580) = 99.50$ ,  $p < 0.001$ . The adjusted model includes all covariates listed in Methods.

**Figure 2.** *Adjusted proportional odds ratios for household food-security status*



### **Diagnostics and sensitivity findings**

The maximum coefficient-level weighted VIF was 2.17, providing no evidence of problematic multicollinearity. In threshold-specific models, associations remained positive and statistically significant for all three lower-food-security categories. For very low food security, adjusted odds ratios ranged from 1.69 at the threshold separating excellent health from all worse categories to 4.19 at the threshold separating poor health from all better categories. This variation indicates that the common POR of 2.60 should be interpreted as an average cumulative association rather than a perfectly constant effect across all health thresholds. Full estimates are provided in Supplementary Table S3.

# Discussion

## Principal findings

In a recent, nationally representative sample of U.S. adults, lower household food security was strongly associated with worse self-rated general health. The pattern was graded: adjusted odds were higher for marginal, low, and very low food security than for high food security, and the largest common association was observed for very low food security. The weighted descriptive results were consistent with the model: fair or poor health was nearly three times as prevalent in the very-low-food-security group as in the high-food-security group.

Adjustment reduced the very-low-food-security POR from 3.31 to 2.60, indicating that measured differences in age, education, smoking, activity, body mass index, sleep, insurance, and other characteristics explained part, but not all, of the observed association. The remaining association should not be interpreted as causal; unmeasured socioeconomic conditions, chronic illness, or reverse directionality may contribute.

## Interpretation in context

The findings align with prior evidence that adults experiencing food insecurity report poorer general, physical, and mental health [4–7]. Several pathways are plausible. Constrained food resources can reduce dietary quality and complicate chronic-disease management, while uncertainty about obtaining food can increase psychological stress. Food insecurity may also reflect broader economic hardship that affects housing stability, access to care, transportation, and medication use. Conversely, poor health can limit employment and increase expenses, making food insecurity more likely. The cross-sectional NHIS data cannot distinguish these directions.

The threshold-specific models add an important qualification. The association was present at every cumulative cut point, but it was stronger for more severe health thresholds. Thus, the central conclusion—a robust association between lower food security and worse health—is stable, while the precise common POR should be treated as a summary across nonidentical threshold effects.

## Strengths

- Use of a large national survey designed to represent the civilian, noninstitutionalized U.S. adult population.
- Correct incorporation of the final annual weight, strata, and primary sampling units for point and variance estimation.
- Retention of the ordered information in four food-security levels and five health levels.
- One common analytic cohort across descriptive and regression results, with transparent missingness reporting.
- Explicit checks of threshold-specific estimates and weighted collinearity, plus a reproducible coding workflow.

## Limitations

- Cross-sectional timing prevents causal or temporal conclusions and permits reverse causation.
- Food security and general health were self-reported and may be affected by recall or reporting differences.
- Complete-case analysis assumes that exclusion related to missing data does not materially bias estimates. Although 92.6% of the sample was retained, this assumption cannot be verified from observed data alone.
- Family income was not included because rigorous use of NHIS income requires the separate multiply imputed income file and combination rules. Education and insurance do not fully capture economic resources, so residual socioeconomic confounding remains likely.
- The activity measure was a simple indicator of any moderate or vigorous leisure-time activity; it did not represent guideline-concordant duration or occupational activity.
- Some race and ethnicity categories were combined to avoid sparse estimates, which can conceal heterogeneity. Race and ethnicity are social classifications rather than biological risk factors.
- The proportional-odds effect was not constant across thresholds. The common POR is an average association, and threshold-specific estimates should be consulted for severe outcomes.
- NHIS excludes institutionalized populations and had a 47.9% Sample Adult response rate in 2024. Survey weighting addresses known selection and nonresponse patterns but cannot eliminate all nonresponse bias.

## Implications

The results support continued attention to food access as part of adult health surveillance. In clinical and community settings, validated food-insecurity screening can identify material needs that may complicate chronic-disease management. Future work should use longitudinal data, incorporate the multiply imputed NHIS income file, examine effect modification, and consider flexible ordinal models that relax the proportional-odds assumption.

## Conclusion

Among U.S. adults in the 2024 NHIS, progressively lower household food security was associated with progressively worse self-rated general health. Adults with very low food security had 2.60 times the adjusted cumulative odds of worse health compared with adults with high food security. The association was robust in direction across health thresholds, although its magnitude was larger at more severe cut points. These findings describe a strong population-level association and should not be interpreted as evidence of causation.

## Data ethics and availability

This analysis used publicly available, deidentified data and did not involve interaction with participants or access to identifiable private information. The 2024 NHIS public-use files and documentation are available from the NCHS website. The raw CSV should be downloaded directly from NCHS rather than committed to a public repository. Repository users remain responsible for complying with applicable institutional requirements and the NHIS data-use guidance.

**Official 2024 NHIS data page:** [cdc.gov/nchs/nhis/documentation/2024-nhis.html](https://cdc.gov/nchs/nhis/documentation/2024-nhis.html)

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# Supplementary Appendix

The appendix records the exact variable definitions, full adjusted model, proportional-odds sensitivity results, and missingness audit used to produce the portfolio report.

**Table S1.** Analysis variables, coding rules, and reference categories

| Construct          | NHIS variable         | Analysis coding                                                             | Reference          |
|--------------------|-----------------------|-----------------------------------------------------------------------------|--------------------|
| Self-rated health  | PHSTAT_A              | Excellent, very good, good, fair, poor; ordered best to worst               | Ordinal outcome    |
| Food security      | FDSCAT4_A             | High, marginal, low, very low; past 30 days                                 | High               |
| Age                | AGEP_A                | 18–34, 35–49, 50–64, 65 or older                                            | 18–34              |
| Sex                | SEX_A                 | Male, female                                                                | Male               |
| Race and ethnicity | HISPALLP_A            | NH White; NH Black; Hispanic; NH Asian; NH AIAN/other/multiple              | NH White           |
| Education          | EDUCP_A               | Bachelor's+; some college/associate; high school/GED; less than high school | Bachelor's+        |
| Marital status     | MARITAL_A             | Married; living with partner; neither                                       | Married            |
| Region             | REGION                | Northeast, Midwest, South, West                                             | Northeast          |
| Smoking            | SMKCIGST_A            | Never, former, current                                                      | Never              |
| Alcohol            | DRKSTAT_A             | Lifetime abstainer; former (2–4); current (5–9)                             | Lifetime abstainer |
| Physical activity  | VIGNR_A, MODNR_A      | Any vs no moderate/vigorous leisure-time activity                           | No activity        |
| Body mass index    | BMICAT_A              | Healthy weight, underweight, overweight, obesity                            | Healthy weight     |
| Sleep duration     | SLPHOURS_A            | 7–9 hours, <7 hours, >9 hours                                               | 7–9 hours          |
| Insurance          | NOTCOV_A              | Covered, not covered                                                        | Covered            |
| Survey design      | WTFA_A, PSTRAAT, PPSU | Final annual weight, pseudo-stratum, pseudo-PSU                             | Not applicable     |

**Note.** NH = non-Hispanic; AIAN = American Indian or Alaska Native; PSU = primary sampling unit.

**Table S2.** Full survey-weighted adjusted proportional-odds model

| Predictor and category         | Adjusted POR (95% CI) | p value |
|--------------------------------|-----------------------|---------|
| <b>Household food security</b> |                       |         |
| High food security             | Reference             | —       |
| Marginal food security         | 1.89 (1.70–2.11)      | <0.001  |
| Low food security              | 1.98 (1.73–2.26)      | <0.001  |
| Very low food security         | 2.60 (2.22–3.04)      | <0.001  |
| <b>Age group</b>               |                       |         |
| 18–34                          | Reference             | —       |
| 35–49                          | 1.43 (1.32–1.55)      | <0.001  |
| 50–64                          | 2.07 (1.91–2.24)      | <0.001  |
| 65 or older                    | 3.03 (2.79–3.30)      | <0.001  |
| <b>Sex</b>                     |                       |         |
| Male                           | Reference             | —       |
| Female                         | 1.04 (0.99–1.10)      | 0.121   |

| Predictor and category                  | Adjusted POR (95% CI) | p value |
|-----------------------------------------|-----------------------|---------|
| <b>Race and ethnicity</b>               |                       |         |
| Non-Hispanic White                      | Reference             | —       |
| Non-Hispanic Black                      | 1.10 (0.99–1.22)      | 0.084   |
| Hispanic                                | 1.08 (0.98–1.18)      | 0.110   |
| Non-Hispanic Asian                      | 1.47 (1.31–1.66)      | <0.001  |
| Non-Hispanic AIAN, other, or multiple   | 1.34 (1.14–1.58)      | <0.001  |
| <b>Educational attainment</b>           |                       |         |
| Bachelor's degree or higher             | Reference             | —       |
| Some college or associate degree        | 1.30 (1.22–1.38)      | <0.001  |
| High school graduate or GED             | 1.50 (1.39–1.62)      | <0.001  |
| Less than high school                   | 2.10 (1.86–2.38)      | <0.001  |
| <b>Marital status</b>                   |                       |         |
| Married                                 | Reference             | —       |
| Living with partner                     | 1.14 (1.03–1.26)      | 0.013   |
| Neither married nor living with partner | 1.21 (1.14–1.29)      | <0.001  |
| <b>Region</b>                           |                       |         |
| Northeast                               | Reference             | —       |
| Midwest                                 | 1.01 (0.93–1.10)      | 0.837   |
| South                                   | 0.97 (0.90–1.04)      | 0.399   |
| West                                    | 1.05 (0.97–1.13)      | 0.203   |
| <b>Smoking status</b>                   |                       |         |
| Never smoker                            | Reference             | —       |
| Former smoker                           | 1.31 (1.24–1.39)      | <0.001  |
| Current smoker                          | 1.80 (1.64–1.98)      | <0.001  |
| <b>Alcohol status</b>                   |                       |         |
| Lifetime abstainer                      | Reference             | —       |
| Former drinker                          | 1.32 (1.18–1.47)      | <0.001  |
| Current drinker                         | 0.94 (0.85–1.03)      | 0.197   |
| <b>Physical activity</b>                |                       |         |
| No moderate/vigorous activity           | Reference             | —       |
| Any moderate/vigorous activity          | 0.55 (0.51–0.58)      | <0.001  |
| <b>Body mass index</b>                  |                       |         |
| Healthy weight                          | Reference             | —       |
| Underweight                             | 1.37 (1.06–1.77)      | 0.016   |
| Overweight                              | 1.33 (1.25–1.42)      | <0.001  |
| Obesity                                 | 2.56 (2.38–2.75)      | <0.001  |
| <b>Sleep duration</b>                   |                       |         |
| 7–9 hours                               | Reference             | —       |
| Less than 7 hours                       | 1.36 (1.28–1.45)      | <0.001  |
| More than 9 hours                       | 2.68 (2.26–3.18)      | <0.001  |
| <b>Health insurance</b>                 |                       |         |
| Covered                                 | Reference             | —       |
| Not covered                             | 0.84 (0.74–0.94)      | 0.003   |

**Note.** POR > 1 indicates higher cumulative odds of worse self-rated health. CI = confidence interval.

**Table S3.** *Threshold-specific adjusted food-security associations*

| Cumulative outcome threshold         | Marginal         | Low              | Very low         |
|--------------------------------------|------------------|------------------|------------------|
| Very good or worse vs excellent      | 1.81 (1.52–2.15) | 1.51 (1.23–1.86) | 1.69 (1.29–2.20) |
| Good or worse vs excellent/very good | 1.90 (1.67–2.15) | 1.84 (1.59–2.14) | 2.24 (1.87–2.67) |

| Cumulative outcome threshold             | Marginal         | Low              | Very low         |
|------------------------------------------|------------------|------------------|------------------|
| Fair or poor vs excellent/very good/good | 1.99 (1.70–2.32) | 2.41 (2.05–2.84) | 3.20 (2.65–3.87) |
| Poor vs excellent/very good/good/fair    | 2.22 (1.71–2.89) | 2.38 (1.80–3.16) | 4.19 (3.10–5.65) |

**Note.** Cells contain adjusted odds ratios (95% CI) from separate survey-weighted binary logistic models. Reference = high food security. All displayed p values are <0.001.

**Table S4.** *Variable-specific missingness before complete-case restriction*

| Variable           | Missing n | Missing % |
|--------------------|-----------|-----------|
| Self-rated health  | 16        | 0.0       |
| Food security      | 1,351     | 4.1       |
| Age                | 52        | 0.2       |
| Sex                | 5         | 0.0       |
| Race and ethnicity | 0         | 0.0       |
| Education          | 147       | 0.5       |
| Marital status     | 1,198     | 3.7       |
| Region             | 0         | 0.0       |
| Smoking status     | 753       | 2.3       |
| Alcohol status     | 877       | 2.7       |
| Physical activity  | 908       | 2.8       |
| BMI category       | 600       | 1.8       |
| Sleep duration     | 1,126     | 3.5       |
| Insurance coverage | 93        | 0.3       |

**Note.** Counts are not mutually exclusive. Complete-case exclusions totaled 2,399; 30,230 of 32,629 adults were retained.